UNIT 7

INBORN ERRORS OF METABOLISM



Sir Archibald Garrod

Sir Archibald Garrod, an English physician, introduced the term "Inborn Errors of Metabolism" in 1908. He postulated that inherited disorders result from reduced activity or complete absence of an enzymes involved in biochemical pathways, causing a block in the metabolic pathway. His study of alkaptonuria led to the finding that a mutation in a gene resulted in the accumulation of homogentisic acid and the presence of black urine on exposure to air. Following this, in 1941, George Beadle and Edward Tatum, using the bread mould *Neurospora,* confirmed Garrod's hypothesis and proposed the 'One gene one enzyme' theory. This hypothesis was further modified in 1957 by Vernom Ingram, as 'One gene one polypeptide' since many proteins are made up of more than one polypeptide chain.

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Learning Objectives

After studying this unit you will be able to understand the following:

- Biochemical basis of Inborn errors of metabolism
- Types of inborn errors of metabolism
- Causes and symptoms of Galactosemia
- Causative factors of Von Gierke disease
- Various types of haemophilia and clotting factors
- Ocular and oculo-cutaneous albinism
- Causes and symptoms of Alkaptonuria
- Causative factors of Tay – Sachs disease.

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